Digital Signal Processing

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FFT algorithm for time-based extraction

- Principle of FFT algorithm for time-based extraction
- Flow chart of FFT algorithm for time-based extraction
- Flow graph characteristics of FFT algorithm for time-based extraction
- The complexity of the FFT algorithm for time-based extraction

FFT algorithm for time-based extraction

$$X[m]=\sum_{k=0}^{N-1}x[k]W_N^{km} \qquad x[k]\rightarrow\begin{cases}x_1[k]=x[2k]\\ x_2[k]=x[2k+1]\end{cases} \qquad k=0,1,\cdots,\frac{N}{2}-1$$

$$=\sum_{k=0}^{N/2-1}x[2k]W_N^{2km}+\sum_{k=0}^{N/2-1}x[2k+1]W_N^{(2k+1)m} \qquad N=2^M$$

$$=\sum_{k=0}^{N/2-1}x[2k]W_{N/2}^{km}+W_N^m\sum_{k=0}^{N/2-1}x[2k+1]W_{N/2}^{km}$$

记 $X_1[m]=\sum_{k=0}^{N/2-1}x_1[k]W_{N/2}^{km}$

$$m=0,1\cdots,\frac{N}{2}-1$$

$$X_2[m]=\sum_{k=0}^{N/2-1}x_2[k]W_{N/2}^{km}$$

FFT algorithm for time-based extraction

$$X[m] = \sum_{k=0}^{N/2-1} x_1[k]\, W_{N/2}^{km} + W_N^m \sum_{k=0}^{N/2-1} x_2[k]\, W_{N/2}^{km}$$

Set $X_1[m] = \sum_{k=0}^{N/2-1} x_1[k] W_{N/2}^{km}$

$$X_2[m] = \sum_{k=0}^{N/2-1} x_2[k] W_{N/2}^{km}$$

$$m = 0,1\cdots,\frac{N}{2}-1$$

Therefore:

$$X[m] = X_1[m] + W_N^m X_2[m] \qquad m = 0,1\cdots,\frac{N}{2}-1$$

$$X[m] = X_1[m] + W_N^m X_2[m] \qquad m = 0,1\cdots,\frac{N}{2}-1$$

Add $N/2$ to m and substitute it into

$$X[m+N/2] = X_1[m+N/2] + W_N^{m+N/2} X_2[m+N/2]$$

Due to the periodicity inherent in $X_1[m]$ and $X_2[m]$, it can be concluded that

$$X[m+N/2] = X_1[m] - W_N^m X_2[m] \quad m = 0,1\cdots,\frac{N}{2}-1$$

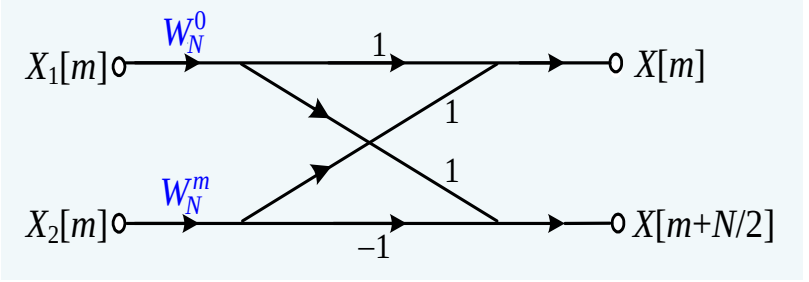
Short sequence DFT combined with long sequence DFT

$$X[m] = X_1[m] + W_N^m X_2[m]$$

$$X[m+N/2] = X_1[m] - W_N^m X_2[m]$$

$$m = 0,1\cdots,\frac{N}{2}-1$$

Butterfly diagram



$$\begin{bmatrix} X[m] \\ X[m+N/2] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} W_N^0 & 0 \\ 0 & W_N^m \end{bmatrix} \begin{bmatrix} X_1[m] \\ X_2[m] \end{bmatrix}$$

FFT algorithm for time-based extraction

- 8-point sequence DFT, can be synthesized from 2 4-point DFTs
- Each 4-point sequence DFT can be synthesized from 2 2-point DFTs
- The FFT algorithm based on time-based extraction is used to transform the time domain to the frequency domain through a 2-point sequence DFT

2-points sequence DFT $x[k]=\{x[0], x[1]\}$

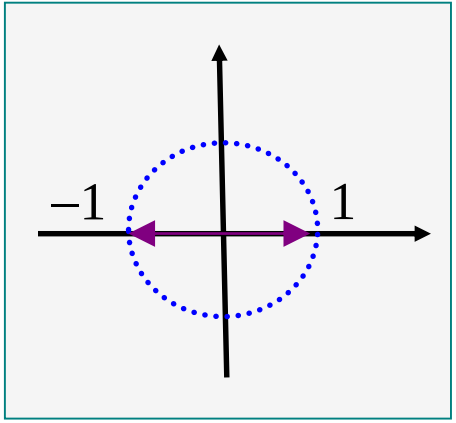
$$X[0] = x[0] + x[1]$$
$$X[1] = x[0] - x[1]$$

$$\begin{bmatrix} X[0] \\ X[1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \end{bmatrix}$$

FFT algorithm for time-based extraction

Short sequence DFT combined with long sequence DFT

$$\begin{bmatrix} X[m] \\ X[m+N/2] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} W_N^0 & 0 \\ 0 & W_N^m \end{bmatrix} \begin{bmatrix} X_1[m] \\ X_2[m] \end{bmatrix}$$

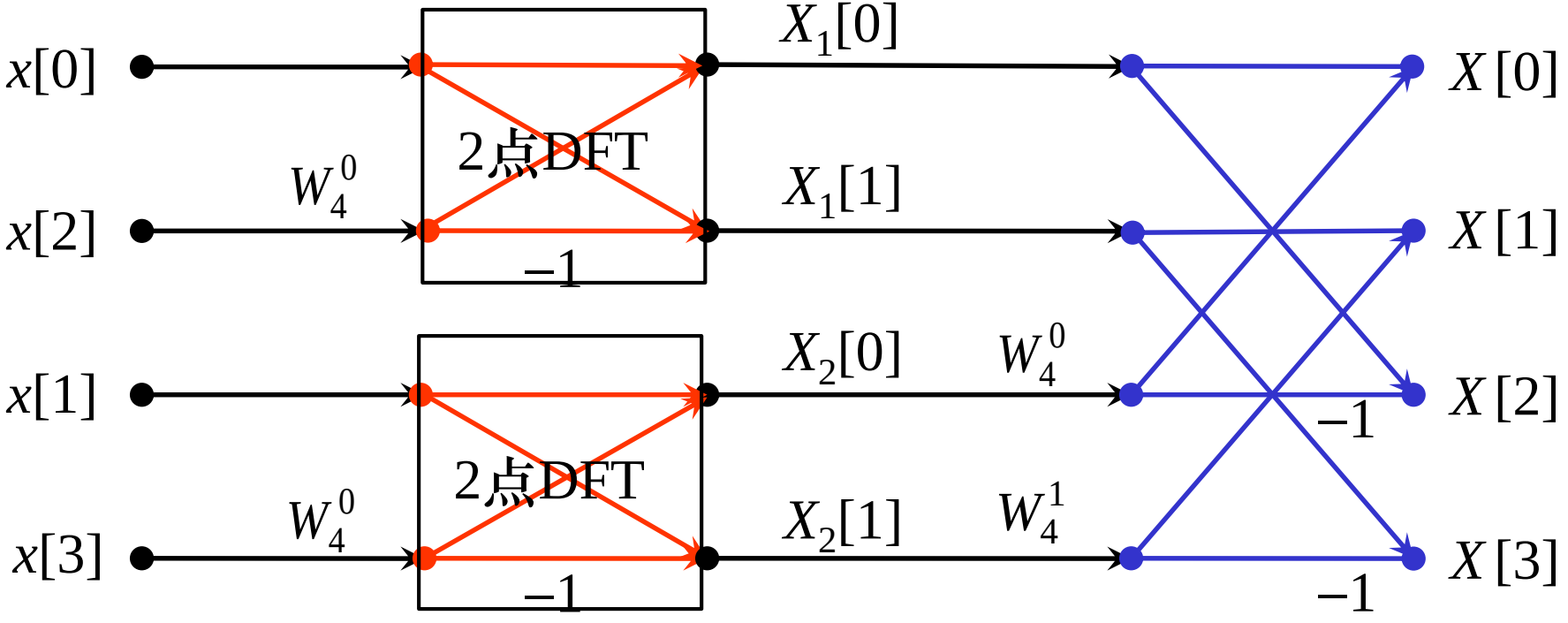


Time domain to frequency domain

$$\begin{bmatrix} X[0] \\ X[1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \end{bmatrix}$$

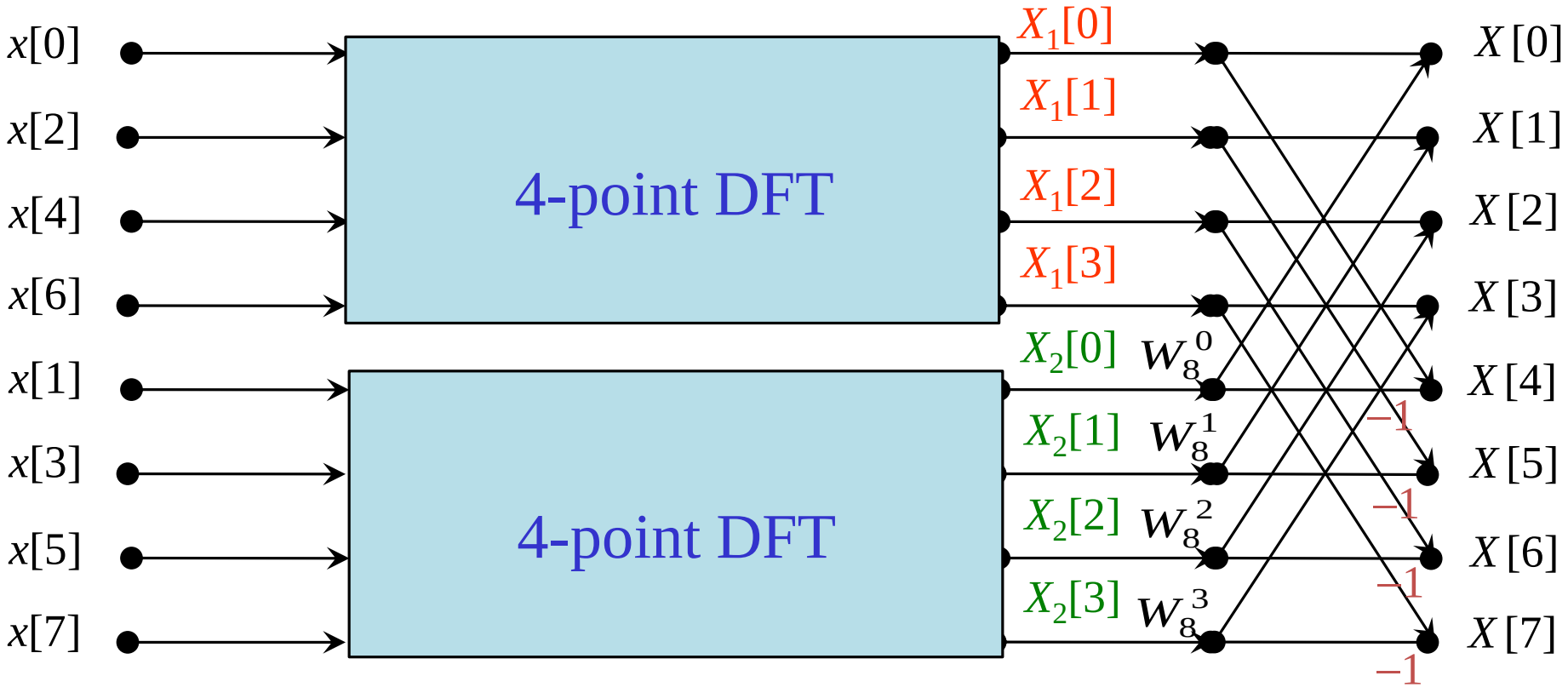
$$X[m]=X_1[m]+W_4^m X_2[m],\quad m=0,1$$

$$X[m+2]=X_1[m]-W_4^m X_2[m],\quad m=0,1$$

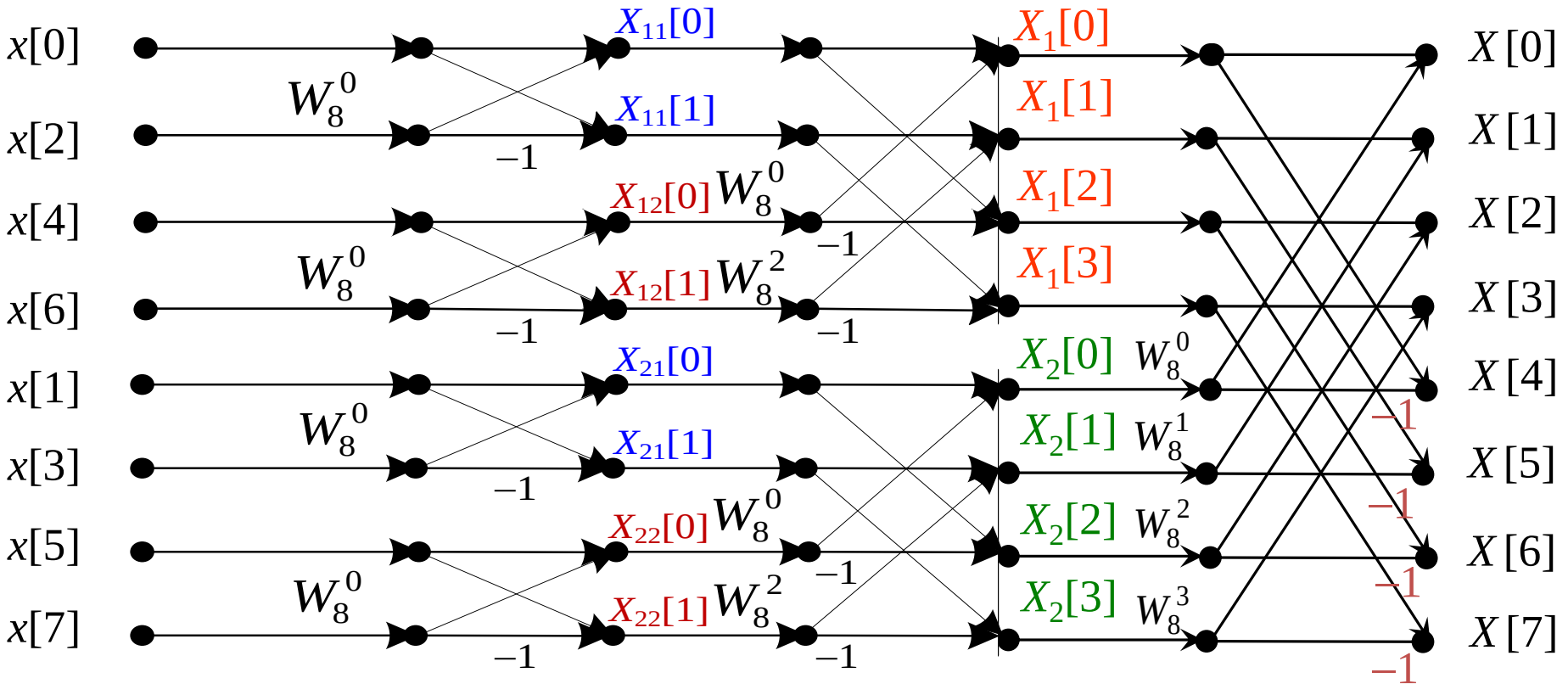


$$X[m] = X_1[m] + W_8^m X_2[m], \quad m = 0,1,2,3$$
$$X[m+4] = X_1[m] - W_8^m X_2[m], \quad m = 0,1,2,3$$

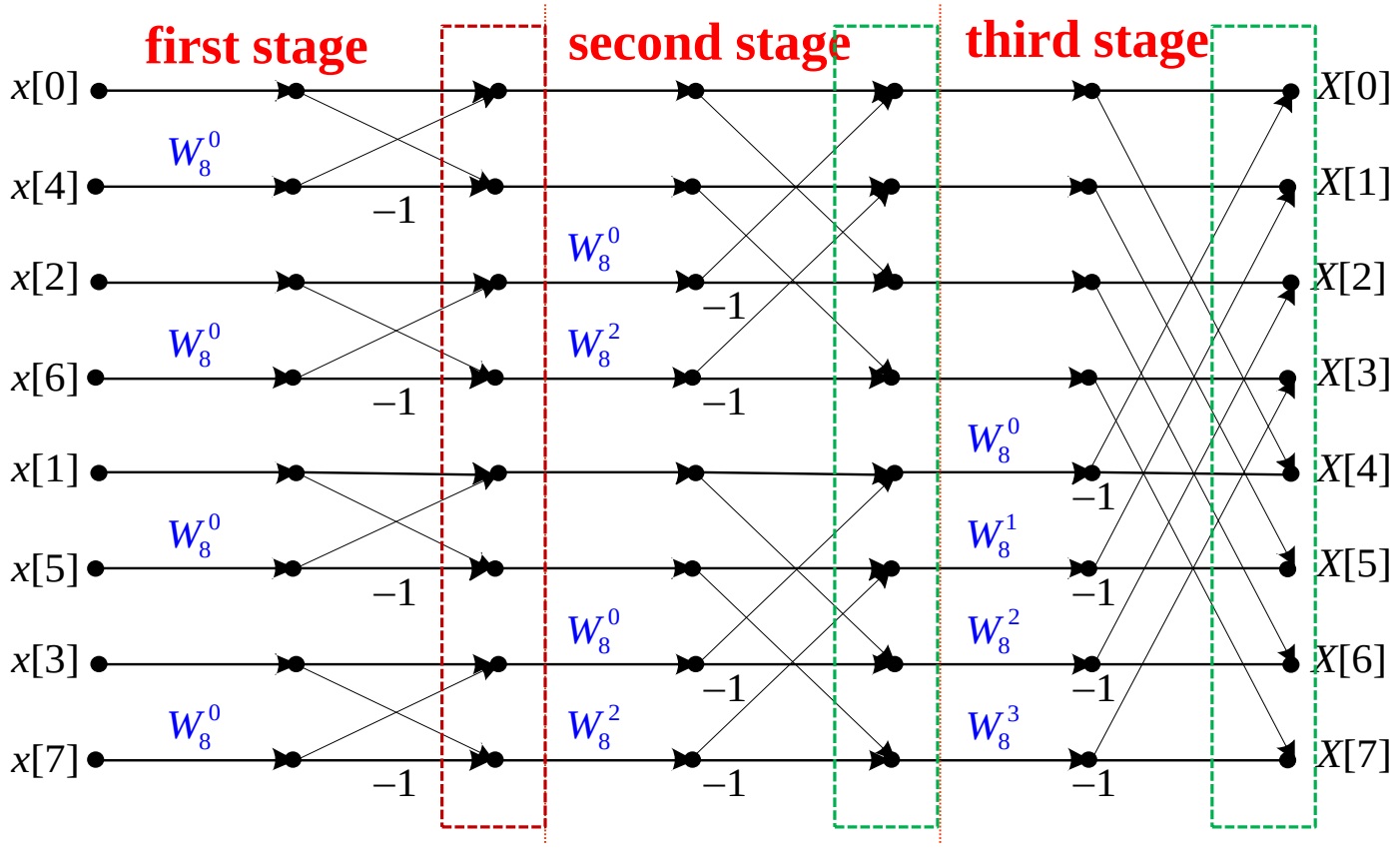
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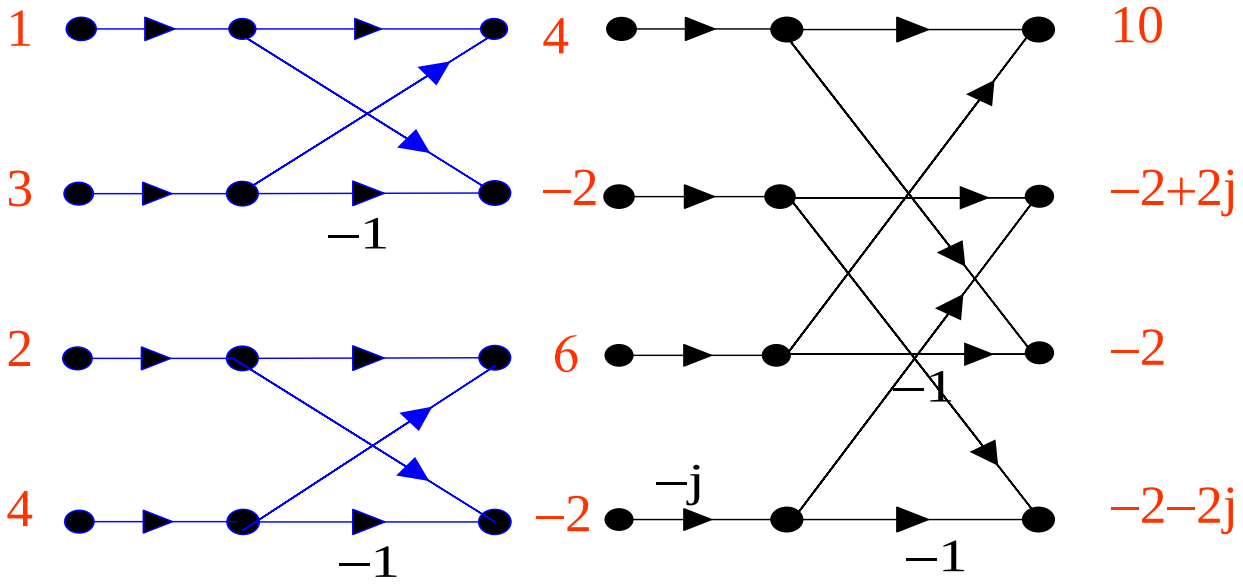
Flow chart of FFT algorithm for time-based extraction



Flow chart of FFT algorithm for time-based extraction

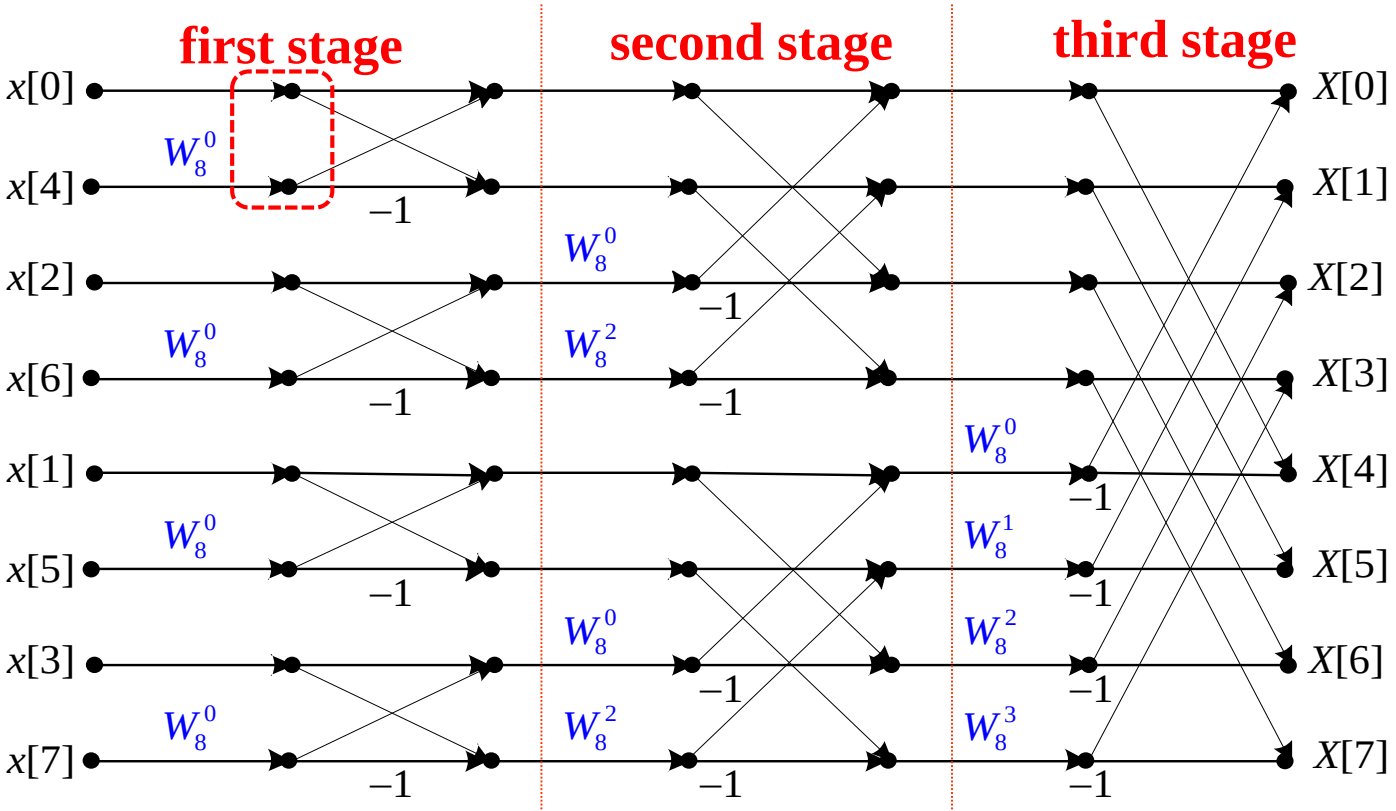


E.g: Known $x[k]=\{1,2,3,4\}$, Using the radix-FFT algorithm for flow graph calculation $X[m] = \mathrm{DFT}\{x[k]\}$。

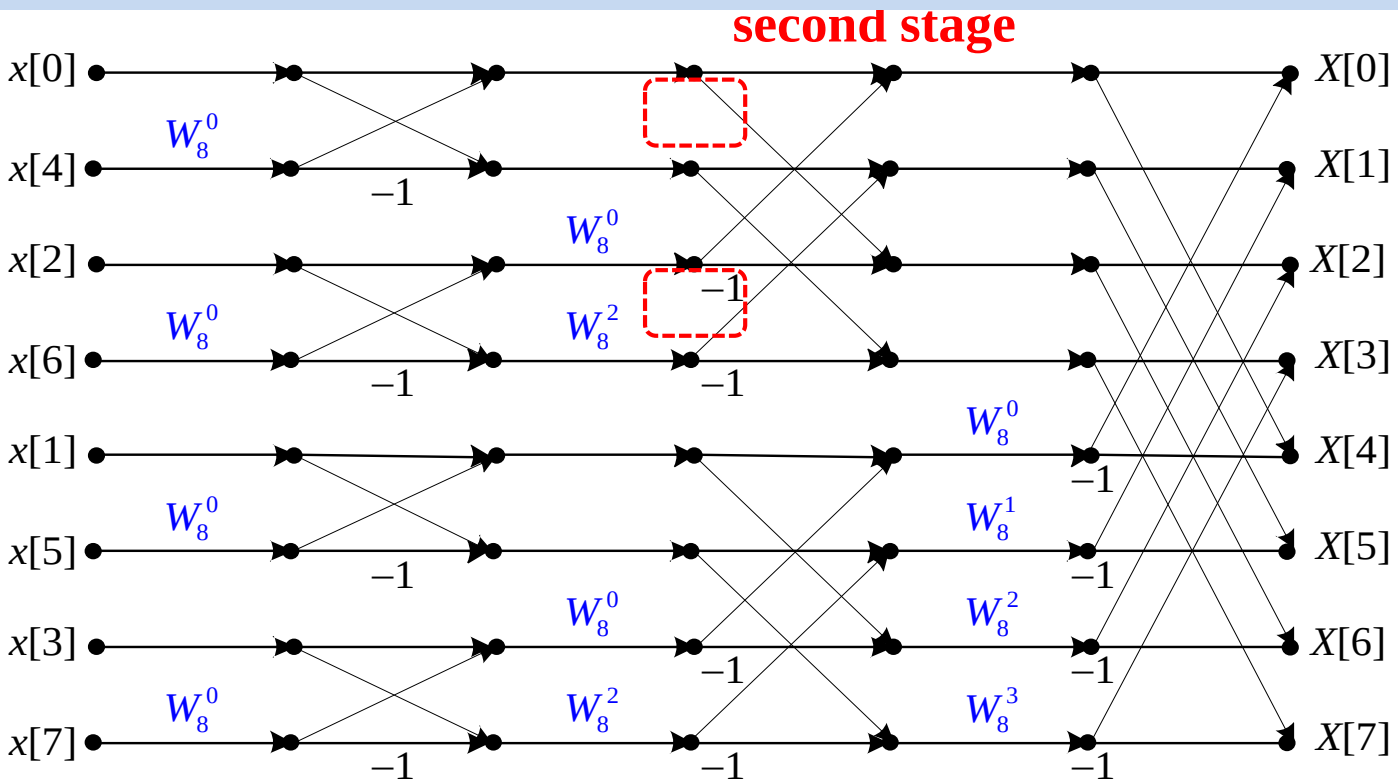


$$\mathrm{DFT}\{x[k]\} = \{10, -2+2j, -2, -2-2j\}$$

twiddle factor



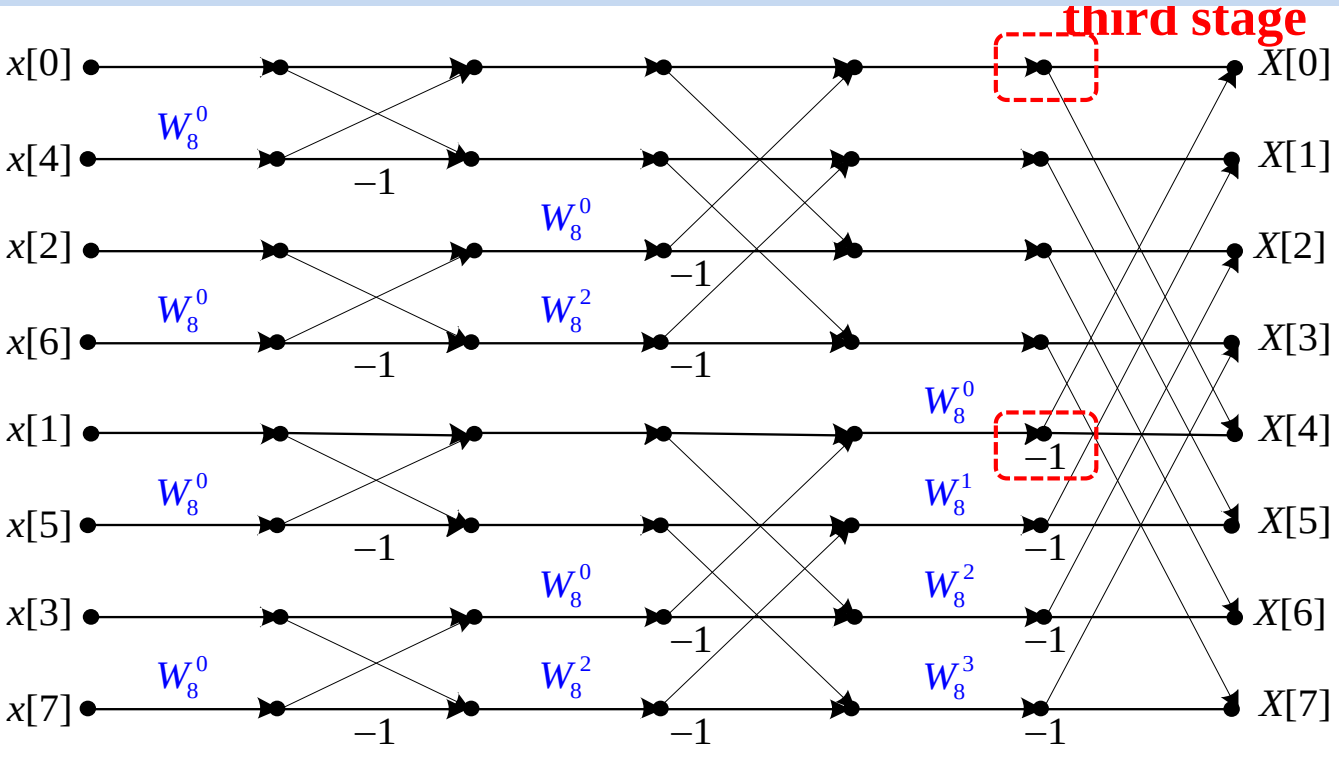
The butterfly shaped coefficients of the first level are all W_N^0, **The distance of the butterfly node is 1.**



twiddle factor

The butterfly coefficient of the second level is W_N^0, $W_N^{N/4}$ The distance between butterfly nodes is 2.

twiddle factor



The butterfly coefficient of the third level is

$$W_N^0,\ W_N^{N/8},\ W_N^{2N/8},\ W_N^{3N/8}$$

The distance between butterfly nodes is 4.

Twiddle factor:

The butterfly shaped coefficients of the first level are all W_N^0 , The distance between butterfly nodes is 1.

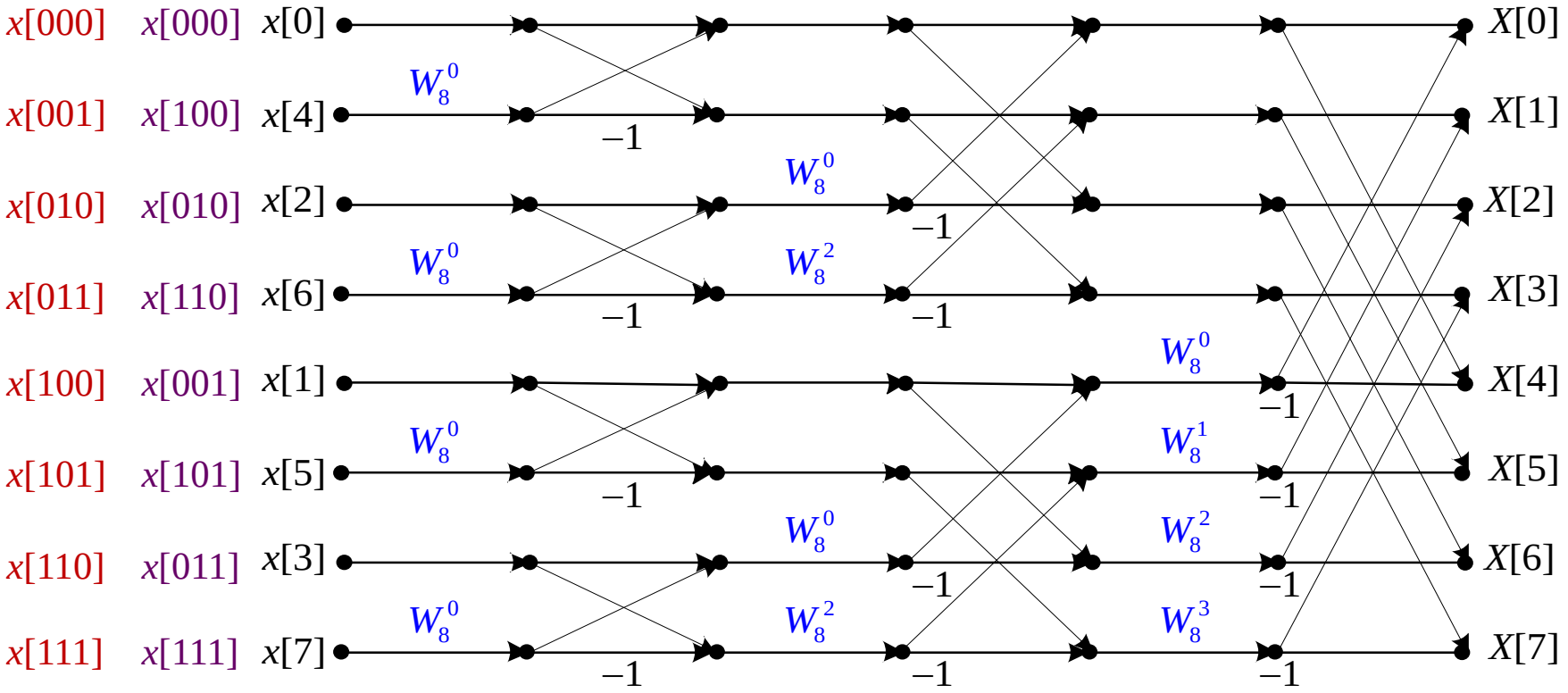
The butterfly coefficient of the second level is $W_N^0, W_N^{N/4}$ The distance between butterfly nodes is 2

The butterfly coefficient of the third level is $W_N^0, W_N^{N/8}, W_N^{2N/8}, W_N^{3N/8}$The distance between butterfly nodes is 4

⋮ ⋮

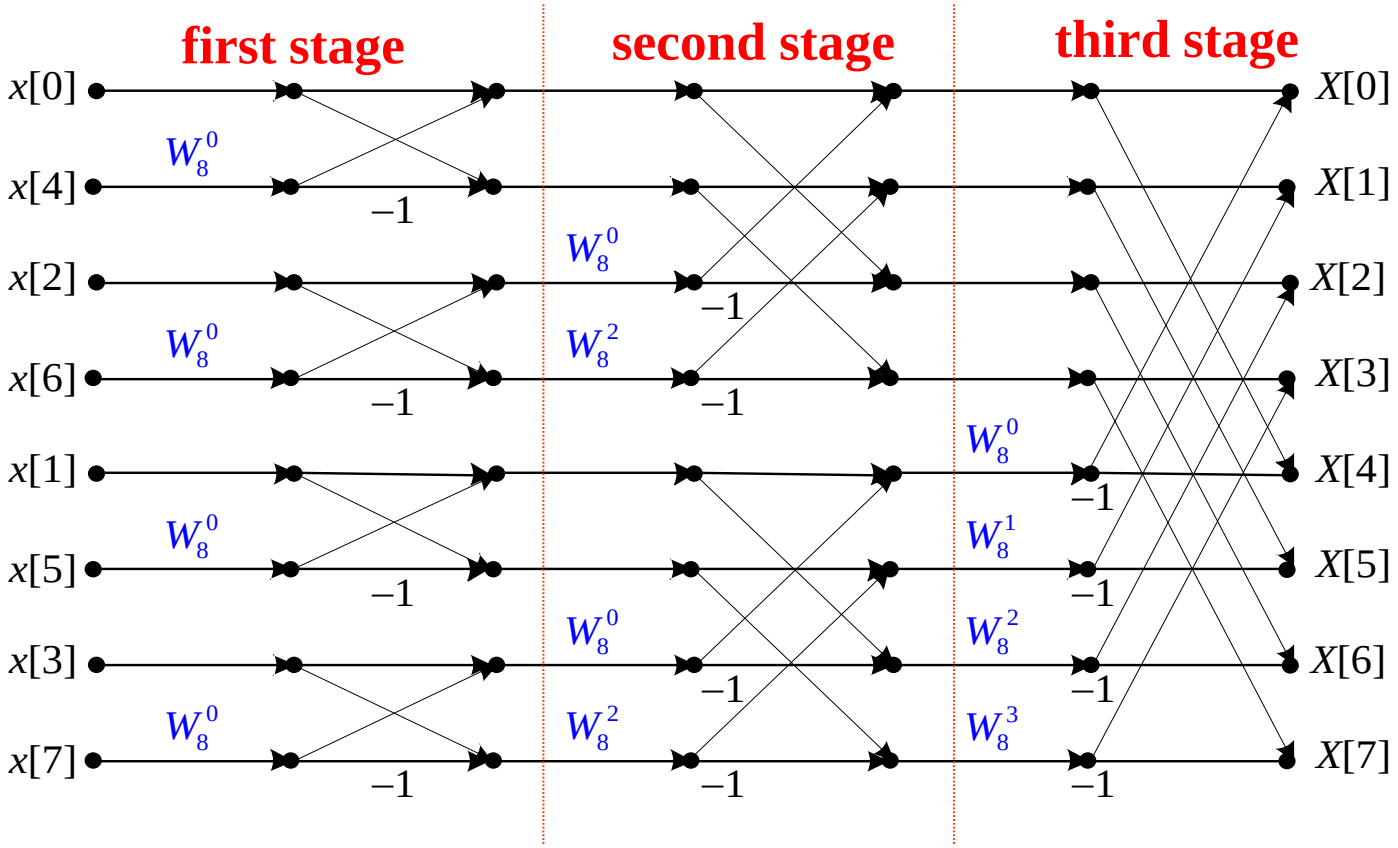
The butterfly coefficient of the *M*-th level is $W_N^0, W_N^1, \cdots, W_N^{(N/2-1)}$The distance between butterfly nodes is *N*/2.

Flow graph characteristics of FFT algorithm for time-based extraction



Flow graph characteristics of FFT algorithm for time-based extraction

In-situ computation

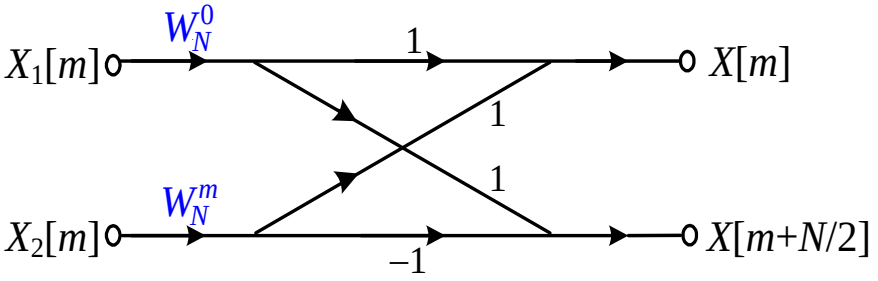


In-situ operation:

In situ operation refers to the process where, after data is input into memory, the results of each level of operation are still stored in their original positions until the final output, without the need to store intermediate calculation results.

In the FFT operation with N=8, the input sequences $x[0]$, $x[4]$, $x[2]$, $x[6]$,..., $x[7]$ can be stored in 8 storage units of $A(0)$, $A(1)$, $A(2)$, $A(3)$,..., $A(7)$, respectively.

The complexity of the FFT algorithm for time-based extraction



$$X[m] = X_1[m] + W_N^m X_2[m]$$

$$X[m+N/2] = X_1[m] - W_N^m X_2[m]$$

Each butterfly operation requires a complex multiplication and a quadratic complex addition,

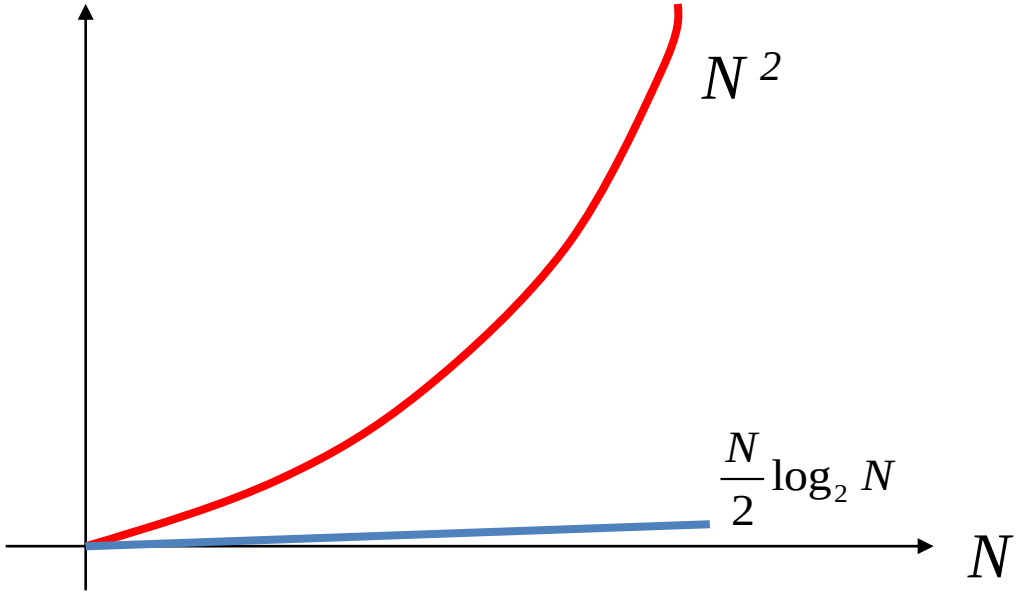
When $N=2^M$, the decomposition level is M, and each level has $N/2$ butterfly shapes.

Complex multiplication: $\frac{N}{2} \times M = \frac{N}{2} \log_2 N$

Complex addition: $\frac{N}{2} \times 2 \times M = N \log_2 N$

The complexity of the FFT algorithm for time-based extraction

Multiple multiplication times



The complexity of the FFT algorithm for time-based extraction

Number of sequence points	DFT		FFT		DFT/FFT	
N	Complex multiplication	Complex addition	Complex multiplication	Complex addition	Complex multiplication	Complex addition
8	64	56	12	24	5.3	2.3
16	256	240	32	64	8.0	3.75
32	1024	992	80	160	12.8	6.2
64	4096	4032	192	384	21.3	10.5
128	16384	16256	448	896	36.6	18.1
256	65536	65280	1024	2048	64.0	31.9
512	262144	261632	2304	4608	113.8	56.8
1024	1048576	1047552	5120	10240	204.8	102.3
2048	4194304	4192256	11264	22528	372.4	186.1
4096	16777216	16773120	24576	49152	682.7	341.3
8192	67108864	67100672	53248	106496	1260.3	630.0

The complexity of the FFT algorithm for time-based extraction


```
N=1024*4;
x = rand(N,1);
tic;
y=fft(x);
t=toc;
fprintf('\nFFT time =%.6e\n',t) ;
tic;
y=dftmtx(N)*x;
t=toc;
fprintf('DFT time =%.6e\n',t);
```

The complexity of the FFT algorithm for time-based extraction

Number of sequence points N	FFT time (s)	DFT time (s)	FFT*100/DFT
1024*2	1.0e-04	0.137	9.49%
1024*4	1.6e-04	0.505	3.96%
1024*8	2.8e-04	1.902	1.46%
1024*16	5.2e-04	141.91	0.04%

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.